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Problem Set Number: 2

Due Date: 2/10/26

1)

a)

Known: $R = 0.5\,m$ $H = 1m$

$T_s = T_{water} = 50°C$

$T_\infty = 20°C$

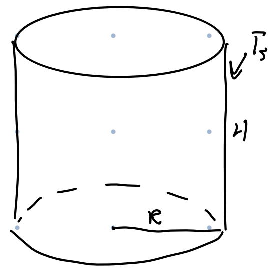
$h = 15 \frac{W}{m^2 \cdot K}$

b) $k = 0.1 \frac{W}{m \cdot K}$

$L = 0.02m$

Find: Electrical power to maintain a constant water temp with and without insulation

Schematic:




Assumptions: Radiation is negligible, steady state, bottom is thermally insulated b) 1D Heat conduction

Analysis:

a)

$\dot{E}_{in} - \dot{E}_{out} + \dot{E}_g = \dot{E}_{st} = 0$

$\cancel{\dot{E}_{in}}^{\,0} + \dot{E}_g = \dot{E}_{out}$

$\dot{E}_g = q = \dot{E}_{out} = hA(T_s - T_\infty)$

$A = 2\pi r h + \pi r^2 = 3.93\, m^2$

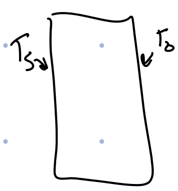
$q = 15 \cdot 3.93 \cdot (50-20) = \boxed{1767\,[W]}$

b) Inner surface

$\dot{E}_{out} = kA \frac{T_s - T_o}{L}$

$\dot{E}_g = q$

$\dot{E}_g = \dot{E}_{out}$




Adding

Outer surface

$\dot{E}_{in} = kA \frac{T_s - T_o}{L}$

$\dot{E}_{out} = hA(T_o - T_\infty)$

$k\cancel{A} \frac{T_s - T_o}{L} = h\cancel{A}(T_o - T_\infty)$

$T_s - T_o = \frac{hL}{k}(T_o - T_\infty)$

$T_o = 27.5°$

$\boxed{q = kA \frac{T_s - T_o}{L} = 442\,[W]}$

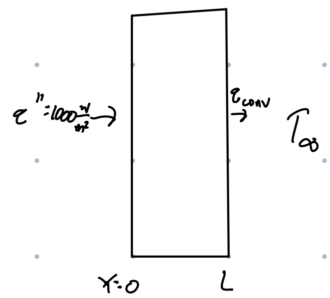
c) The heat needed in part b) is roughly one quarter of that in a) so clearly the insulation makes massive improvements in saving energy.

2) Known: $L = 0.1\,m$ $\quad T_\infty = 20°C$

$k = 1 \frac{W}{m^2 K}$ $\quad h = 100 \frac{W}{m^2 K}$

Find: $T_{x=0}$ when heater supplies $1000\, W/m^2$, $q''$ when $T_{x=0} = 200°C$

Schematic:




Assumptions: Heater is infinitely thin, steady state, neglect radiation, Heater is thermally insulated on left

Analysis:

$$\dot{E}_{in} - \dot{E}_{out} + \cancel{\dot{E}_g}^{\,0} = \dot{E}_{st} = 0$$

$$\dot{E}_{in} = \dot{E}_{out} = q''$$

$x=0$ $\quad\quad x=L$

$$q'' = kA \frac{T_{x=0} - T_{x=L}}{L} \qquad kA\frac{T_{x=0} - T_{x=L}}{L} = hA(T_{x=L} - T_\infty)$$

$$k\cancel{A} \frac{T_0 - T_L}{L} = h\cancel{A}(T_L - T_\infty) = 1000 \frac{W}{m^2}$$

$$\boxed{\begin{aligned} T_L &= 30°C \\ T_0 &= 130°C \end{aligned}}$$

b)

$$q'' = k\frac{200 - T_L}{L} = h(T_L - T_\infty)$$

$$200 = \frac{Lh}{k}(T_L - T_\infty) + T_L$$

$$200 + (10 \cdot T_\infty) = 11 T_L$$

$$T_L = 36.4°$$

$$\boxed{q'' = 1636 \frac{W}{m^2}}$$

c) If we make the plastic layer thinner, the heat will leave more quickly, allowing higher heat flux.

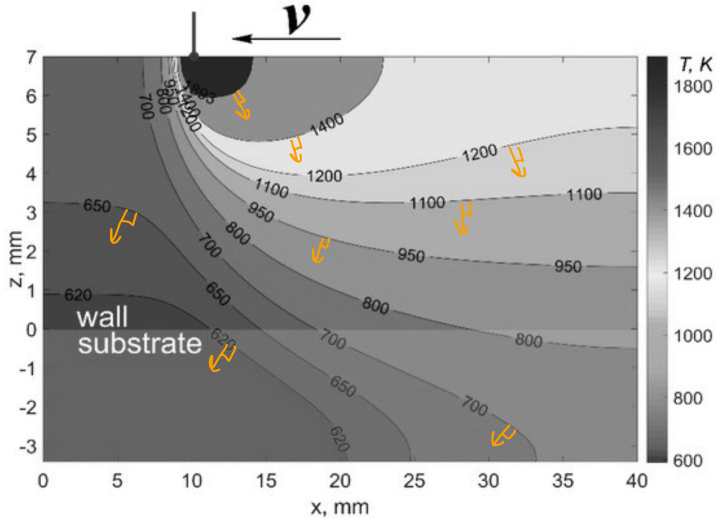
Similarly, if we change the material to one of a higher k, then more heat can be acceptable.

Finally, if we add holes to the surface, increasing surface area, more heat will be dissipated due to convection.

3) Known: Temp of isotherms on x-z plot.

Find: Draw heat flux vectors

Schematic:




Assumptions: 2D heat transfer

a) Analysis → on figure

Vectors should be normal to isotherms, pointing from hotter to colder temps.

b) Left of the laser spot has higher heat flux because the temperature drops off more quickly on the left, meaning greater heat losses in a smaller section.